Introduction to Transparent Bayesian Data Analysis

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Research interests

Data visualization, visual communication
Predictive modeling using statistics and machine learning
Model explanations

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Research interests

Data visualization
Tools for transparent statistical analysis
Predictive models: user modeling, model explanation

Why use **Bayesian data analysis?**

What is **Bayesian data analysis?**

How to do **Bayesian data analysis?**

What to report as **a transparent practice?**

Limitations of **Bayesian data analysis**

Fumeng

Abhraneel

Why use Bayesian data analysis?

What is Bayesian data analysis?

How to do Bayesian data analysis?

What to report as a transparent practice?

Limitations of Bayesian data analysis

Fumeng

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Why use Bayesian data analysis?

Why use Bayesian data analysis?

We can specify models based on the data generation process

Allows us to incorporate prior beliefs and domain knowledge

Analysis model directly corresponds to experimental design.

We can have estimates of uncertainty for each parameter

Each model parameter is a random variable and has a distribution.
Provides distributional information about parameters (e.g., beyond mean, like standard deviation).

Why use Bayesian data analysis?

NHST is evaluating how extreme the data is given the null hypothesis

$$P(data|hypothesis) < 0.05$$

… which is the opposite of answering the research question

$$P(hypothesis|data) \propto P(data|hypothesis) \times P(hypothesis)$$

Bayesian data analysis is evaluating the hypothesis given data.

Why use transparent Bayesian data analysis?

Transparency What you should consider disclosing to be more transparent about your study and analysis

Why BDA Our highlights of the advantages of using Bayesian data analysis

Questions?

What is Bayesian data analysis?

Bayes rule

Likelihood for θ. Conditional probability of Y given θ.

Prior probability distribution for unknown θ.

$$P(\theta \mid Y) = \frac{P(Y \mid \theta)\, P(\theta)}{P(Y)}$$

Y, i.e., the observed data.
θ, e.g., the distribution mean.

Posterior probability distribution for unknown θ given Y. Capture what we think about θ given what we know about Y.

Normalize the distribution
P(Y) does not vary with θ .
We will eventually discard it.

Bayes rule

Likelihood for θ. Conditional probability of Y given θ.

Prior probability distribution for unknown θ.

$$P(\theta \mid Y) \propto P(Y \mid \theta)\, P(\theta)$$

Y, i.e., the observed data.
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Posterior probability distribution for unknown θ given Y. Capture what we think about θ given what we know about Y.

Bayesian *t*-test

The example dataset comes from Yvonne Jansen and Kasper Hornbæk.

```
pose_df = readr::read_csv("data/poses_data.csv", show_col_types = FALSE) %>%
  mutate(condition) %>%
   group_by(participant)
```

(output)

participant <dbl>	condition <chr>	change <dbl>
1	expansive	3.362832
2	constrictive	29.147982
3	expansive	25.409836
4	constrictive	54.069767
5	expansive	-36.644592
6	constrictive	29.756098

The data has been aggregated for each participant.

The independent variables (IVs)
condition = **expansive** indicates expansive posture
condition = **constrictive** indicates constrictive posture.

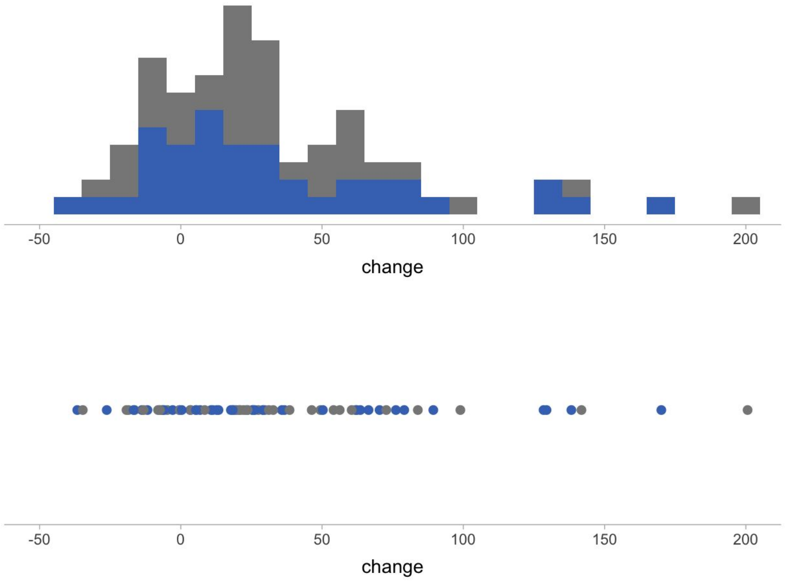
The **dependent variable (DV)** is **change**, the change of a behavior.

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Plot the data

```
wrap_plots(
  pose_df %>% ... geom_histogram(aes(color = condition))+ ...,
  pose_df %>% ... geom_point(aes(color = condition))+ ...
nrow = 2)
```

(output)

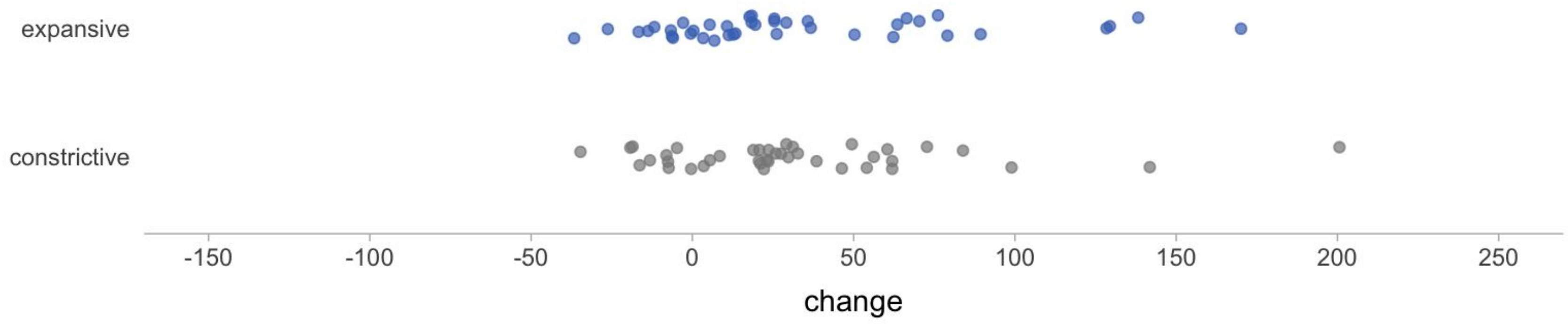


This dataset describes the change of a risk behavior in two conditions: **expansive** and **constrictive.**

We are interested in the **differences** between the two conditions.

Bayesian intuition

Suppose we are interested in **differences in *mean*** and let's focus on **expansive** first.

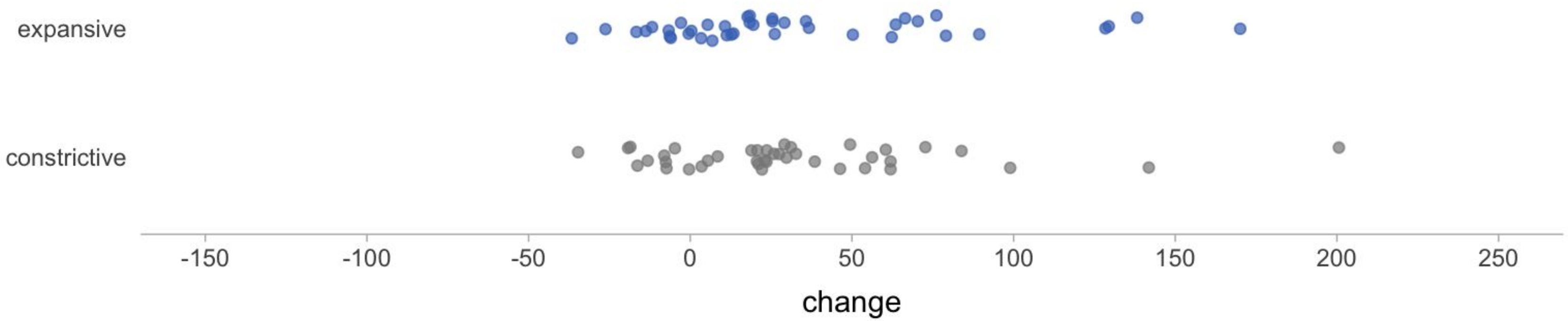


Bayesian intuition

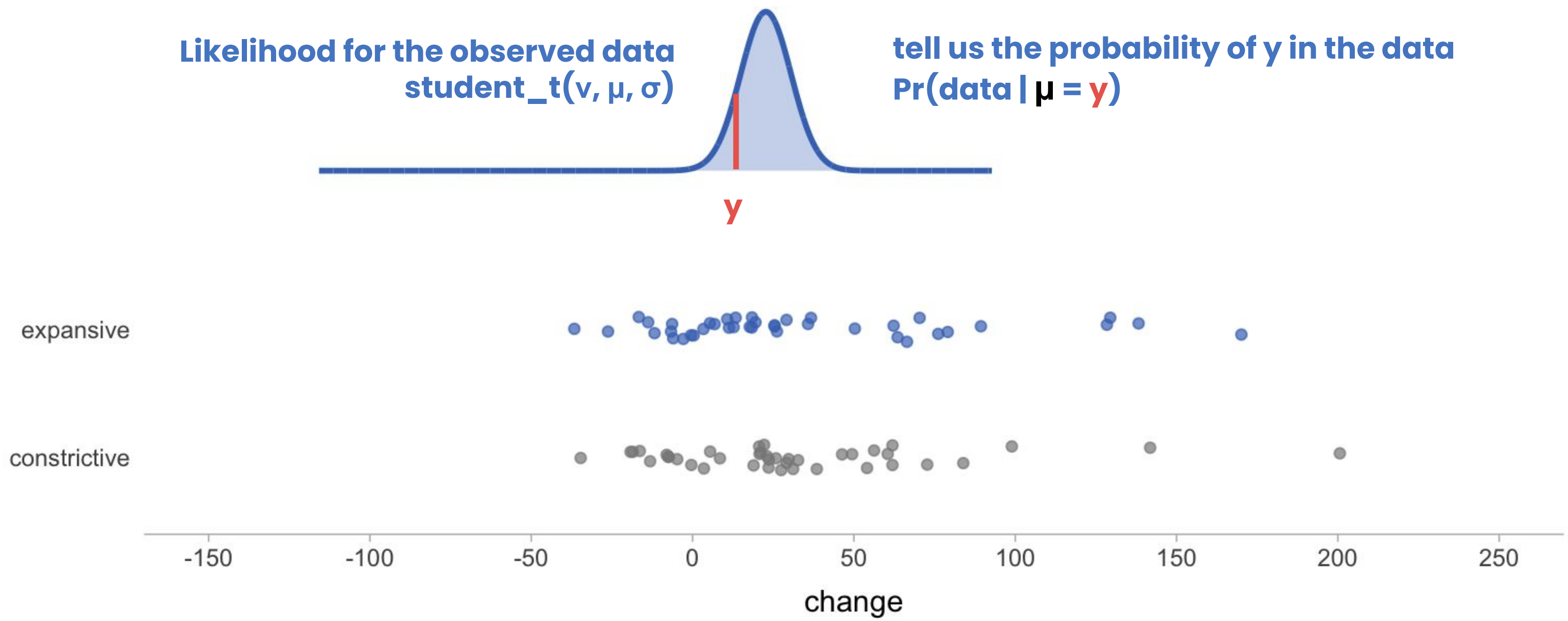
The Naive prior belief – using 0 to indicate that we think the change is around **0**.

Prior distribution
Normal(0, σ)

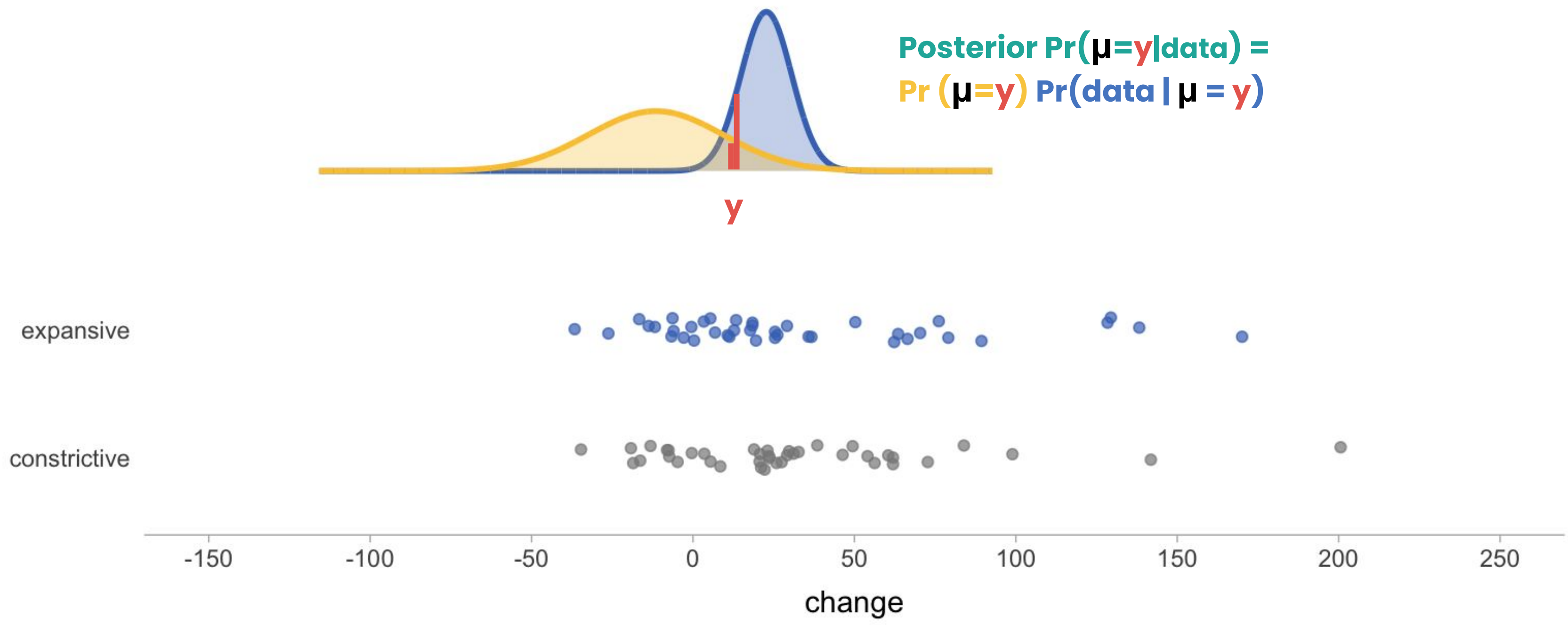
Prior Probability
Pr (μ=y)



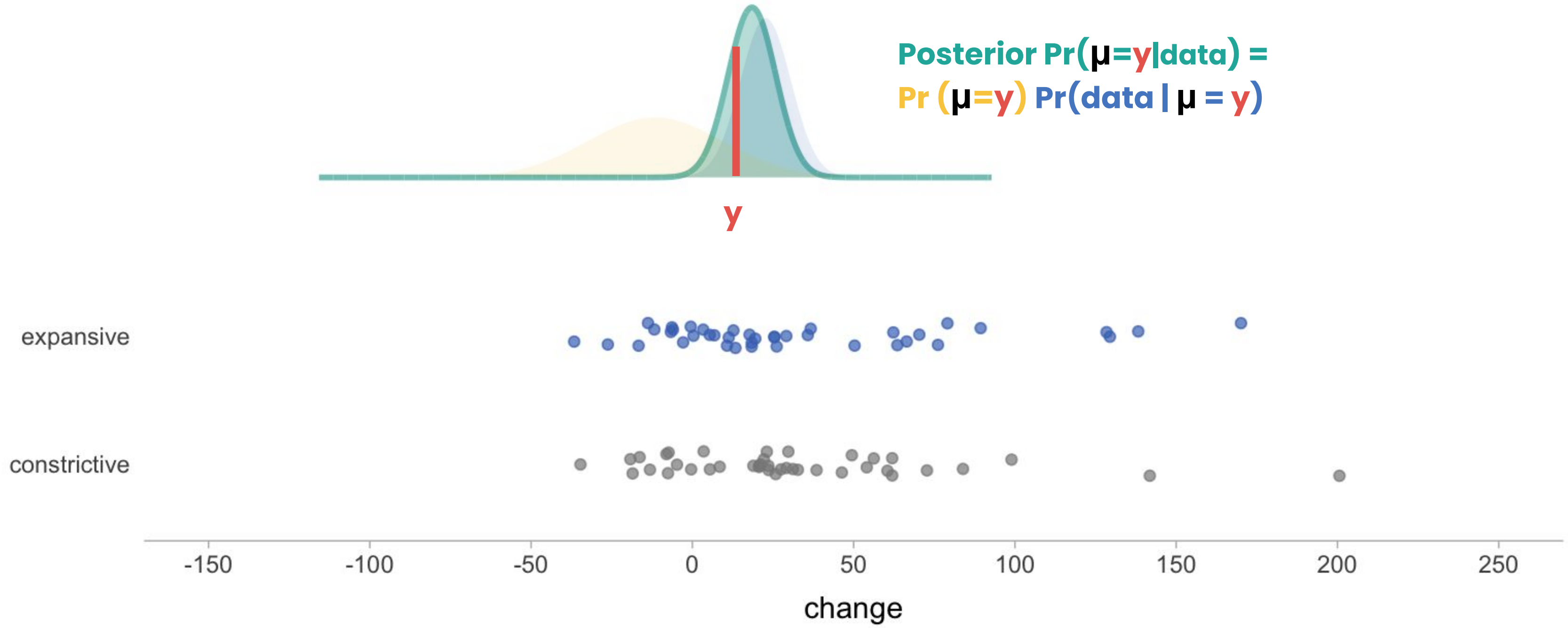
Bayesian intuition



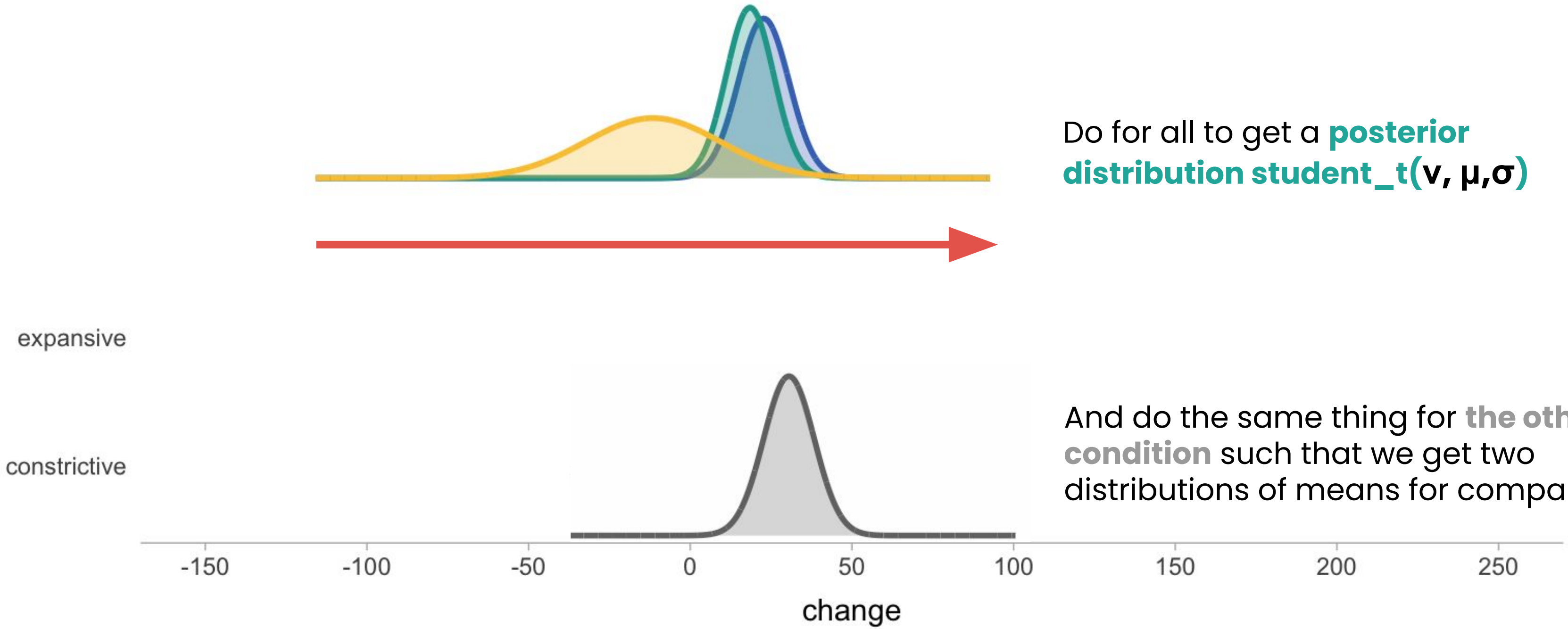
Bayesian intuition



Bayesian intuition



Bayesian intuition

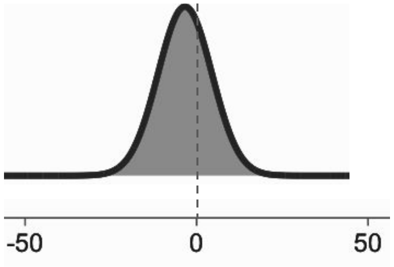


Do for all to get a **posterior distribution student_t(v, μ,σ)**

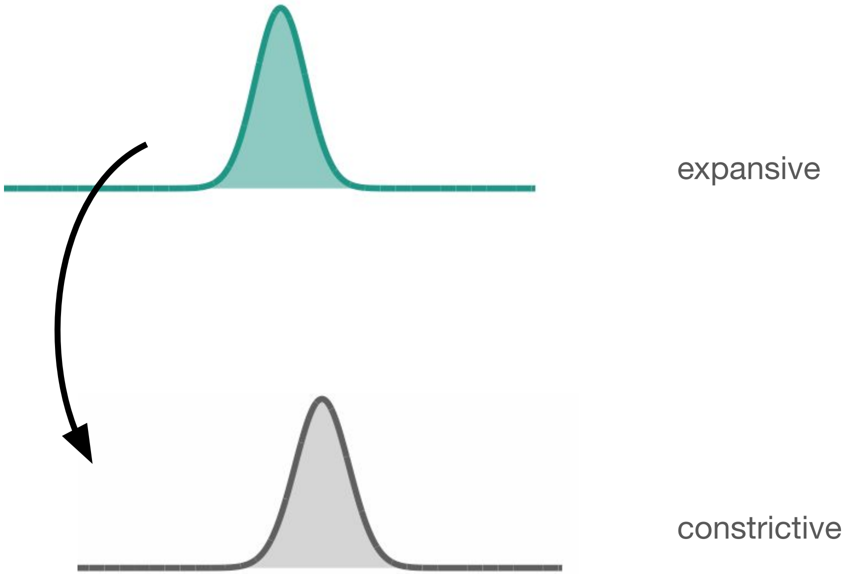
And do the same thing for **the other condition** such that we get two distributions of means for comparison

Bayesian intuition

We **subtract** posterior draws to get **a distribution of difference in mean** and compare it with 0.



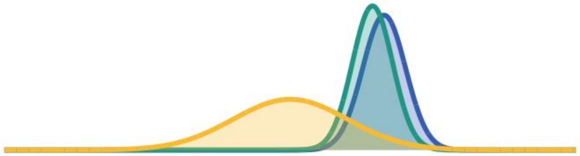
expansive - constrictive



Questions?

How to do Bayesian data analysis?

Bayesian data analysis

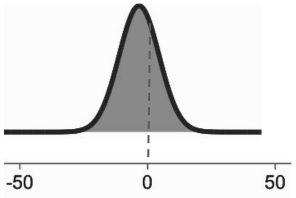


Three steps

1. **Define the model**: the model formula (e.g., how DV varies with IVs), **the distribution of the observed data** (likelihood), and **priors**.

2. Conditioned on the observed data, **update the model** with the information contained in the data using a Markov Chain Monte Carlo (MCMC) process.

3. **Evaluate** the MCMC process, model fit, **posterior predictions** etc.



Bayesian data analysis

Stan

A platform for statistical modeling and probabilistic programming.

An engine for building and executing Markov Chain Monte Carlo processes.

Stan interfaces with the most popular data analysis languages (**R**, Python, shell, MATLAB, Julia, Stata).

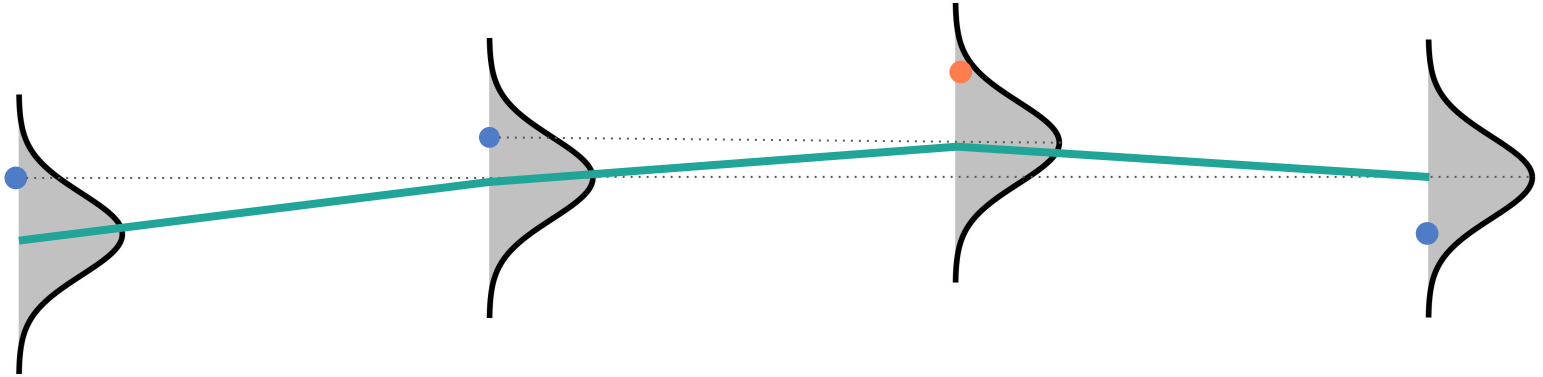
Markov Chain Monte Carlo (MCMC)

It is not always possible to mathematically compute posterior probability distributions.

We use a Markov Chain Monte Carlo (MCMC) algorithm to draw samples from an unknown and complex target distribution (the posterior distribution).

Markov Chain Monte Carlo (MCMC)

After thousands of steps, we have MCMC traces for means. We will emphasize these traces later in this lecture.



Draw from the proposed $N(\mu_0, \sigma)$

Posterior $Pr(\mu_1|data) > Pr(\mu_0|data)$?
accept

Using the last value as the new mean $N(\mu_2, \sigma)$
-> new draw

Posterior $Pr(\mu_2|data) > Pr(\mu_1|data)$?
accept

Using the last value as the new mean $N(\mu_3, \sigma)$
-> new draw

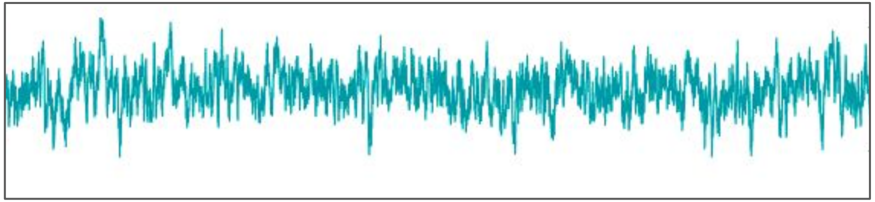
Posterior $Pr(\mu_3|data) > Pr(\mu_2|data)$?
reject (roughly)

Using the previous value as the new mean $N(\mu_2, \sigma)$
-> new draw

repeat

https://www.youtube.com/watch?v=OTO1DygELpY

Markov Chain Monte Carlo (MCMC)



Not apparent problem

High serial correlation

We want to get independent samples.

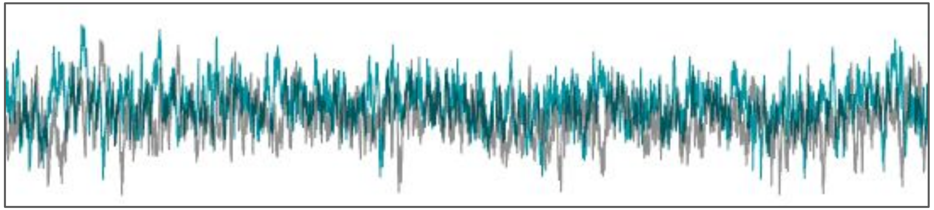
Given by a metric called Effective Sample Size (**ESS**).

It is roughly the number of iterations divided by the step size.

This should be at least a few hundreds and ideally at least 1,000.

https://www.statlect.com/fundamentals-of-statistics/Markov-Chain-Monte-Carlo-diagnostics

Markov Chain Monte Carlo (MCMC)



We want to get independent samples.

Start a different MCMC process, called a "chain"

We want to know if the sampling process converges.

e.g., not all over the place, chains agree with each other

Given by **R-hat** = total variance across chains / variance within chains (roughly)

1 or close to 1 means the model converged.

https://arxiv.org/pdf/1903.08008.pdf Rank-normalization, folding, and localization: An improved Rb for assessing convergence of MCMC

Bayesian data analysis in R

`library(rstan)`	The must-have package, the R interface to Stan. It complies the model (in C++), does sampling, provides diagnostics, etc.

`library(brms)`
`library(rstanarm)`
`library(rethinking)`
	They simplify the model specification, especially if you are familiar with **R**. You can use any of these three. This lecture uses `library(brms)`.

`library(tidybayes)`	It simplifies the draws from posterior distribution.
`library(bayesplot)` `library(ggdist)`	Packages to visualize posterior distributions, model diagnostics, etc. We use `library(bayesplot)` in this lecture. Next lecture will cover `library(ggdist)`.

Resume after 05:00 m

Questions?

Why use Bayesian data analysis?

What is Bayesian data analysis?

How to do Bayesian data analysis?

What to report as a transparent practice?

Limitations of Bayesian data analysis

Fumeng

Abhraneel

Example 1:
Continuous dependent variable

Example 1: Continuous dependent variable

Participants are placed in either a *expansive* condition or *constrictive* condition. These are the aggregated data you saw:

```
read_csv("data/poses_data.csv")
```

participant	condition	change
1	expansive	3.362832
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Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

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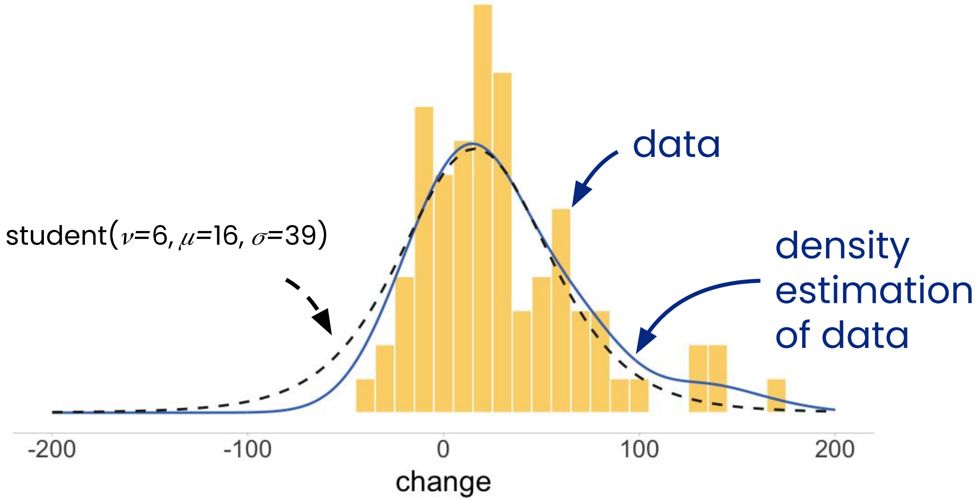

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Example 1: Continuous dependent variable

Step 1: Build the model (likelihood)

We implement the *Bayesian estimation for 2 groups* described by Jansen and Hornbæk. This model estimates mean and standard deviation for each condition.

likelihood $y_i \sim t(\nu, \mu, \sigma)$

```
dataset2.brm.student.formula <-
      bf(
         change ~ condition,
         family = student()
      )
```

Why BDA

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 1: Continuous dependent variable

Step 1: Build the model (priors)

We call **`get_prior`** to see what priors are there.

```
get_prior(change ~ condition, family = student(), data = dataset1)
```

Example 1: Continuous dependent variable

Step 1: Build the model (priors)

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```
get_prior(change ~ condition, family = student(), data = dataset1)
```

prior	class	coef	group	resp	dpar	nlpar	bound	source
flat	b							default
	b	conditionexpansive						default
student_t(3, 22.6, 38.8)	Intercept							default
gamma(2, 0.1)	nu							default
student_t(3, 0, 2.5)	sigma							default

default priors
given by **`brms`**

Example 1: Continuous dependent variable

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default priors given by **brms**

parameters for the likelihood

Example 1: Continuous dependent variable

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student_t(3, 0, 2.5)	sigma							default

default priors given by **`brms`**

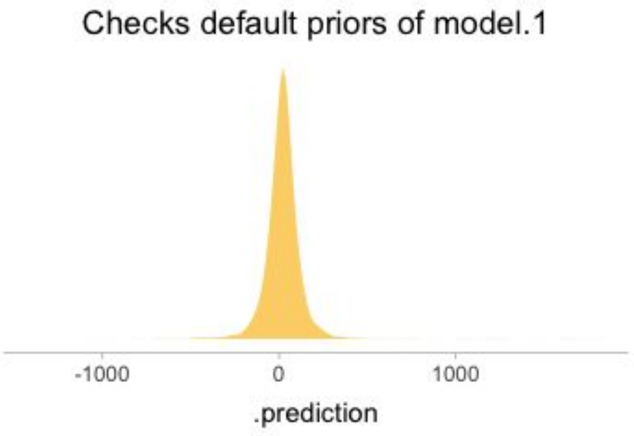
parameters for the likelihood

coefficients for the conditions

Example 1: Continuous dependent variable

Step 1: Build the model (prior predictive checks)

```
brm.student = brm(
 bf(...),
 prior = ...,
 data = ...,
 sample_prior = 'only')
```



Example 1: Continuous dependent variable

Step 1: Build the model (priors)

```
priors <- c(
   prior(normal(0, 2), class = "b"),
   prior(student_t(3, 0, 2.5), class = "sigma"),
   prior(student_t(3, 22.6, 10), class = "Intercept")
)
```

prior	class	coef	group	resp	dpar	nlpar	bound	source
normal(0, 2)	b							default
	b	conditionexpansive						default
student_t(3, 22.6, 10)	Intercept							default
student_t(3, 0, 2.5)	sigma							default
gamma(2, 0.1)	nu							default

Example 1: Continuous dependent variable

Step 1: Build the model (likelihood+priors)

Pass everything to `brm()` and stan. Since we've familiar with the process, we can use a simpler coding style.

```
pose_df.brm.student = brm(
 bf(change ~ condition,
    family = student()),
 prior = c(
   prior(normal(0, 2), class = "b"),
   prior(student_t(3, 0, 2.5), class = "sigma"),
   prior(student_t(3, 0, 5), class = "Intercept")
 ),
 data = pose_df
)
```

formula/likelihood

priors

```
Compiling Stan program...
Start sampling
Running MCMC with 4 sequential chains...
```

Step 2: Computing the posterior probability

Example 1: Continuous dependent variable

Step 3: Evaluate the fit (summary)

```
summary(pose_df.brm.student)
```

```
Family: student
  Links: mu = identity; sigma = identity; nu = identity
Formula: change ~ condition
   Data: pose_df (Number of observations: 80)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Population-Level Effects:
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept             22.84      4.49    14.50    31.77 1.00     2686     2092
conditionexpansive    -0.26      1.97    -4.10     3.62 1.00     2726     2987

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    28.35      4.13    20.76    36.92 1.00     2180     2520
nu        3.68      2.35     1.57     9.04 1.00     2172     2158
```

Example 1: Continuous dependent variable

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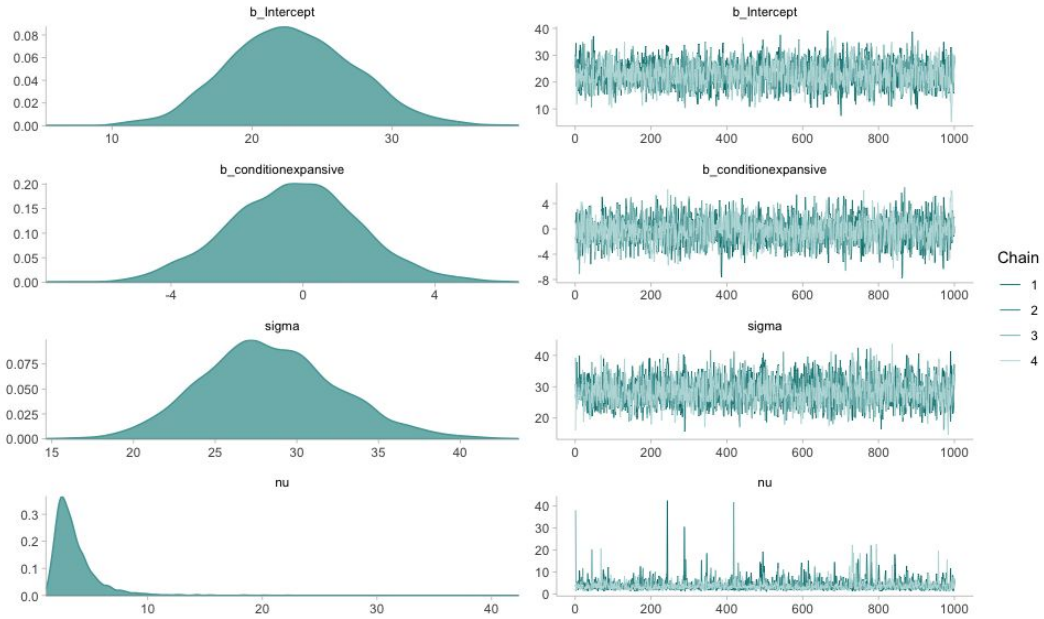
```

Example 1: Continuous dependent variable

Step 3: Evaluate the fit (MCMC traces)

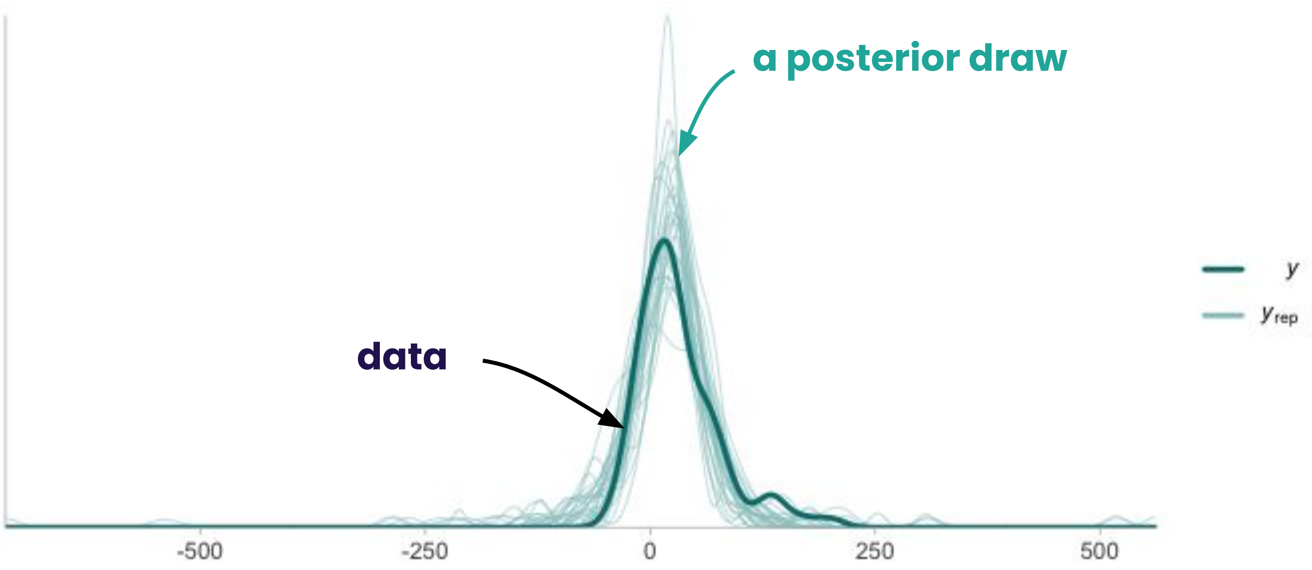
```
plot(pose_df.brm.student)
```



Example 1: Continuous dependent variable

Step 3: Evaluate the posterior predictions (ppc_dens_overlay)

```
ppc_dens_overlay( y = ..., yrep = ... )
```

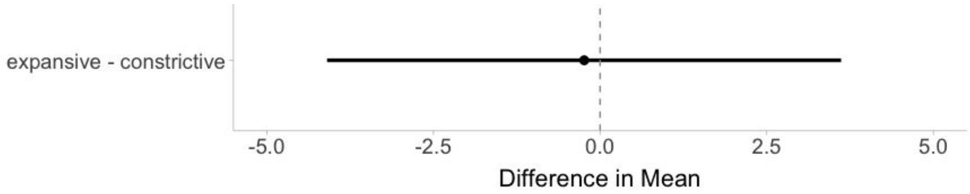


Example 1: Continuous dependent variable

Step 3: Compare the posterior distributions

```
epred_comparison <-
  tibble(condition = c(expansive, constrictive)) %>%
  add_epred_draws(pose_df.brm.student) %>%
  ...
  compare_levels(variable = .epred, by = condition)
```

.draw	condition	diff
1	expansive - constrictive	-1.66
2	expansive - constrictive	-1.068
3	expansive - constrictive	2.55



Questions?

Example 2:
Integer-only dependent variable

Example 2: Integer-only dependent variable

Let's load a new dataset. Participants are placed in either an *expansive* condition or *constrictive* condition.

```
read_csv("data/poses_data_raw.csv")
```

participant	condition	pumps	exploded
1	expansive	62	0
1	expansive	27	1
1	expansive	47	0
1	constrictive	86	1
1	expansive	60	0
…	…	…	…
2	constrictive	4	0

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (likelihood)

We implement the *Bayesian estimation for 2 groups* described by Jansen and Hornbæk. This model estimates mean and standard deviation for each condition.

likelihood $y_i \sim \textit{Neg-Binomial}(\mu, \Phi)$

```
brm.negbinom.formula <-
      bf(
         change ~ condition,
         family = negbinomial()
      )
```

Why BDA

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

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likelihood $y_i \sim \textit{Neg-Binomial}(\mu, \Phi)$

```
brm.negbinom.formula <-
      bf(
         change ~ condition,
         family = negbinomial()
      )
```

Why BDA

Reference: *A student's guide to Bayesian statistics* by Ben Lambert, Table 8.1

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Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

Because we are modeling the data directly, we do not need to look at the data to specify priors

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

Because we are modeling the data directly, we do not need to look at the data to specify priors

We can make intuitive guesses regarding the data distribution

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

Because we are modeling the data directly, we do not need to look at the data to specify priors

We can make intuitive guesses regarding the data distribution

Range of values: [0, 128]

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

Because we are modeling the data directly, we do not need to look at the data to specify priors

We can make intuitive guesses regarding the data distribution

Range of values: [0, 128]
Mean: 64

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

Because we are modeling the data directly, we do not need to look at the data to specify priors

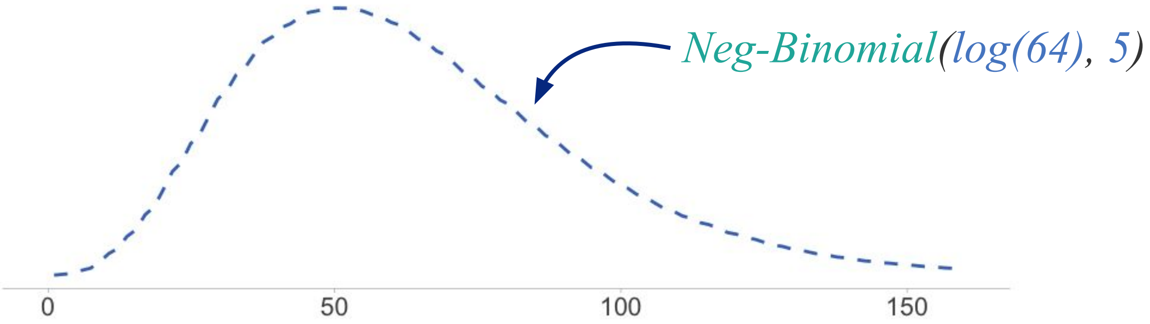
We can make intuitive guesses regarding the data distribution

Range of values: [0, 128]
Mean: 64
Distribution: Negative-Binomial (already assumed)

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

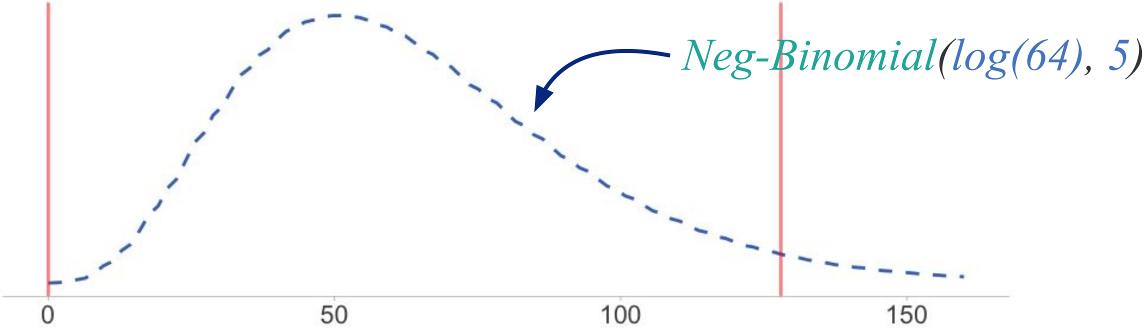
Step 1: Build the model (weakly informative priors)



Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

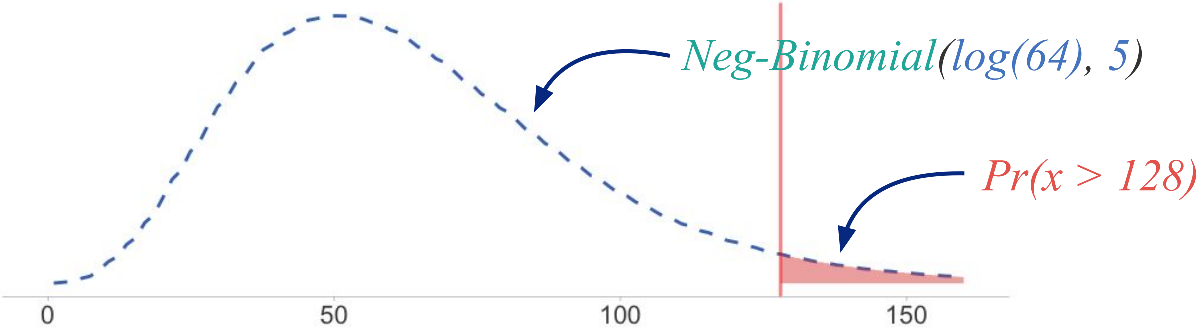
Step 1: Build the model (weakly informative priors)



Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

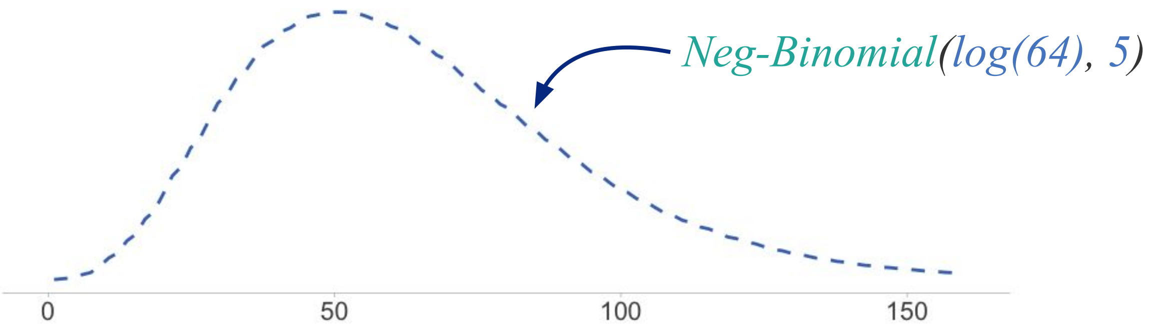


Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Step 1: Build the model (weakly informative priors)

```
priors1 <- c(
   prior(normal(4.1, 0.5), class = Intercept),
   prior(normal(0, 0.5), class = b),
   prior(gamma(5, 1), class = shape)
)
```



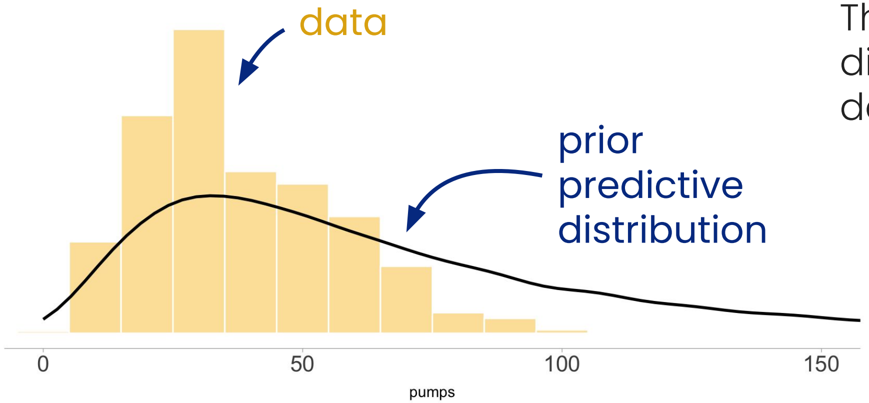
Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Transparency

Example 2: Integer-only dependent variable

Step 1: Build the model (prior predictive checks)

```
brm.negbinom = brm(
 bf(...),
 prior = ...,
 data = ...,
 sample_prior = 'only')
```



The prior predictive density looks not to dissimilar to the actual distribution of the data.

Example 2: Integer-only dependent variable

Step 1: Build the model (likelihood+priors)

Pass everything to `brm()`

```
brm.negbinom = brm(
 bf(change ~ condition,
    family = negbinomial()),
 prior = c(
   prior(normal(4.1, 0.5), class = "Intercept"),
   prior(normal(0, 0.5), class = "b")
   prior(gamma(5, 1), class = "shape")
 ),
 data = pose_raw_df
)
```

formula/likelihood

priors

```
Compiling Stan program...
Start sampling
Running MCMC with 4 sequential chains...
```

Step 2: Computing the posterior probability

Example 2: Integer-only dependent variable

Step 3: Evaluate the fit (summary)

```
summary(brm.negbinom)
```

```
Family: poisson
  Links: mu = log
Formula: pumps ~ 1 + condition
   Data: pose_raw_df (Number of observations: 2400)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

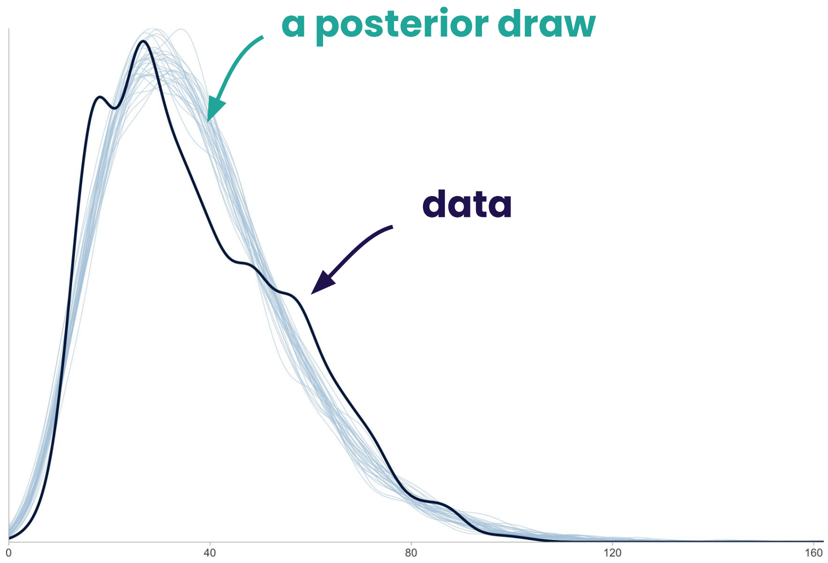
Population-Level Effects:
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept              3.64      0.01     3.61     3.66 1.00    11389     9056
conditionexpansive    -0.01      0.02    -0.05     0.03 1.00    11512     8288

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
shape     4.59      0.15     4.32     4.88 1.00    11208     8380
```

Example 2: Integer-only dependent variable

Step 3: Evaluate the posterior predictions (ppc_dens_overlay)

```
ppc_dens_overlay( y = ..., yrep = ... )
```

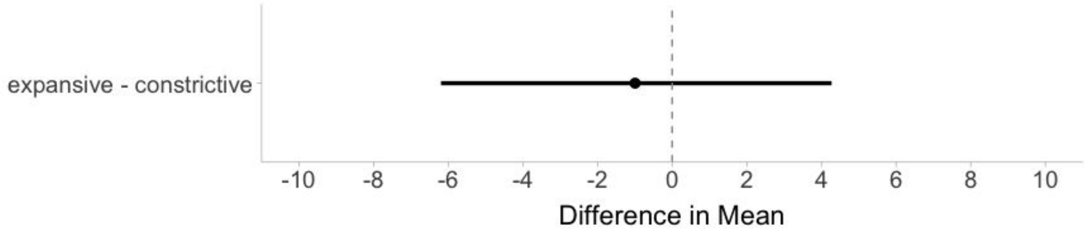


Example 2: Integer-only dependent variable

Step 3: Compare the posterior distributions

```
epred_comparison <-
  tibble(condition = c(expansive, constrictive)) %>%
  add_epred_draws(brm.negbinom) %>%
  ...
  compare_levels(variable = .epred, by = condition)
```

.draw	condition	diff
1	expansive - constrictive	0.3618687
2	expansive - constrictive	1.1572392
3	expansive - constrictive	-0.3027622



Example 2: Integer-only dependent variable

Other effects to consider:

- **Trial:** participants behaviors may change as they repeat trials

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Example 2: Integer-only dependent variable

Other effects to consider:

- **Trial:** participants behaviors may change as they repeat trials
- **Random intercepts for each participant:** Participants may be risk-averse or risk-seeking

Yvonne Jansen and Kasper Hornbæk. "How Relevant are Incidental Power Poses for HCI?." *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Questions?

What to report?

What to report?

Guidelines

Kruschke, John K. "Bayesian analysis reporting guidelines." *Nature human behaviour* 5.10 (2021): 1282-1291.

van Doorn, Johnny, et al. "The JASP guidelines for conducting and reporting a Bayesian analysis." *Psychonomic Bulletin & Review* 28.3 (2021): 813-826.

Table 1 | List of key reporting points for the BARG

Preamble

A. *Why Bayesian.* If the audience requires it, explain what benefits will be gleaned by a Bayesian analysis (as opposed to a frequentist analysis).
B. *Goals of analysis.* Explain the goals of the analysis. This prepares the audience for the type of models to expect and how the results will be described.

Step 1. Explain the model

A. *Data variables.* Explain the dependent (predicted) variables and independent (predictor) variables.
B. *Likelihood function and parameters.* For every model, explain the likelihood function and all the parameters, distinguishing clearly between parameters of primary theoretical interest and ancillary parameters. If the model is multilevel, be sure that the hierarchical structure is clearly explained, along with any covariance structure if multivariate parameter distributions are used.
C. *Prior distribution.* For every model, explain and justify the prior distribution of the parameters in the model.
D. *Formal specification.* Include a formal specification (mathematical or computer code) of the likelihood and prior, located either in the main text or in in publicly and persistently accessible online supplementary material.
E. *Prior predictive check.* Especially when using informed priors but even with broad priors, it is valuable to report a prior predictive check to demonstrate that the prior really generates simulated data consistent with the assumed prior knowledge.

Step 2. Report details of the computation

A. *Software.* Report the software used, including any specific added packages or plugins.
B. *MCMC chain convergence.* Report evidence that the chains have converged, using a convergence statistic such as PSRF, for every parameter or derived value.
C. *MCMC chain resolution.* Report evidence that the chains have high resolution, using the ESS, for every parameter or derived value.
D. *If not MCMC.* If using some computational procedure other than MCMC, be aware of and report inherently inaccurate approximations, especially for the limits of credible intervals.

Step 3. Describe the posterior distribution

A. *Posterior predictive check.* Provide a posterior predictive check to show that the model usefully mimics the data.
B. *Summarize posterior of variables.* For continuous parameters, derived variables and predicted values, report the central tendency and limits of the credible interval. Explicitly state whether you are using density-based values (mode and HDI) or quantile-based values (median and ETI), and state the mass of the credible interval (for example, 95%).
C. *BF and posterior model probabilities.* If conducting model comparison or hypothesis testing, report the BF and posterior probabilities of models for a range of prior model probabilities.

Step 4. Report decisions (if any) and their criteria

A. *Why decisions?* Explain why the decisions are theoretically meaningful and which decision procedure is being used. Regardless of which decision procedure is used, if it addresses null values, it should be able to accept the null value not only reject it.
B. *Loss function.* If utilities and a loss function for a decision rule are defined, these should be explained and reported.
C. *ROPE limits.* If using a continuous-parameter posterior distribution as the basis for decision, state and justify the limits of the ROPE and the required probability mass.
D. *BF, decision threshold and model probabilities.* If using model comparison or hypothesis testing as the basis for a decision, state and justify the decision threshold for the posterior model probability, and the minimum prior model probability that would make the posterior model probability exceed the decision threshold.
E. *Estimated values too.* If deciding about null values, always also report the estimate of the parameter value (central tendency and credible interval).

Step 5. Report sensitivity analysis

A. *For broad priors.* If the prior is intended to be vague or only mildly informed so that it has minimal influence on the posterior, show that other vague priors produce similar posterior results.
B. *For informed priors.* If the prior is informed by previous research, show what posterior results from a vague prior or from a range of differently informed priors.
C. *For default priors.* If using a default prior, show the effect of varying its settings. Be sure that the range of default priors constitutes theoretically meaningful priors, and consider whether they mimic plausible empirically informed priors.
D. *BFs and model probabilites.* If the analysis involves model comparison or hypothesis testing, then for each prior report not only the BFs but also the posterior model probabilities for a range of prior model probabilities.
E. *Decisions.* If making decisions, report whether decisions change under different priors. For BFs, report changes in the minimum prior model probability needed to achieve decisive posterior model probability.

Step 6. Make it reproducible

A. *Software and installation.* Explain all the software that is necessary and where to obtain it. If possible, use non-proprietary software.
B. *Software version details.* The posted script should include detailed information about the software version numbers.
C. *Script and data.* Post the complete analysis script (that is, computer code) and data in a stable public repository with persistent URLs, so that anyone can download it and exactly reproduce the analysis. Be sure that it is clear how to navigate the site and find relevant files, for example, with a wiki overview or readme file. If posting data, be sure that it respects privacy and copyright restrictions. If the original data cannot be posted publicly, it may be helpful to post dummy data of the same form so that users can verify the operation of the analysis script.
D. *Readable for humans.* Make the posted script genuinely readable by human beings. Annotate the code with thorough explanatory comments and spatially arrange the code for human readability.
E. *All auxiliary files.* Check that all the needed auxiliary files (utility scripts, image files, bibliography files, formatting files and so on) are also posted.
F. *Runs as posted.* Check that the posted script and accompanying files run as is when downloaded to a different computer. The code should have no lines that load files from personal computer directories or non-persistent URLs.
G. *MCMC chains for time-intensive runs.* For MCMC runs that take a long time to compute, it is helpful to post an MCMC chain so that people can inspect the MCMC chain without having to wait through an entire run duration.
H. *Reproducible MCMC.* To make MCMC chains exactly reproducible, the pseudo-random number generators should be explicitly seeded.

These points are discussed in the main text, and an extended example is presented in Supplementary Information (also at https://osf.io/w7cph/). When reporting an analysis, the points on this list should be addressed in the main text or in appendices and supplementary material. The specific sequential ordering of points is suggested, and not a requirement. BF, Bayes factor; ESS, effective sample size; ETI, equal-tailed interval; HDI, highest-density interval; MCMC, Markov chain Monte Carlo; PSRF, potential scale reduction factor; ROPE, region of practical equivalence.

What to report?

From our experience

What to report in a transparent practice?

In paper

Step 1. Model specification
Step 2. Core model diagnostics
Step 3. Posterior distributions that answer your research questions

What to report in a transparent practice?

In paper

Step 1. Model specification
Step 2. Core model diagnostics
Step 3. Posterior distributions that answer your research questions

In supplementary materials

Step 1. Model specification details, prior checks, model alternatives
Step 2. Full model diagnostics
Step 3. Full posterior distributions
Other package information, ReadMe

What to report in a transparent practice?

In paper

Step 1. Model specification
Step 2. Core model diagnostics
Step 3. Posterior distributions that answer your research questions

In supplementary materials

Step 1. Model specification details, prior checks, model alternatives
Step 2. Full model diagnostics
Step 3. Full posterior distributions
Other package information, ReadMe

Basically, your data & code

What to pre-register?

In paper

Step 1. Model specification
Step 2. Core model diagnostics
Step 3. Posterior distributions that answer your research questions

Step 1 : Model specification in paper

Option 1. High-level description

> We use a Negative Binomial likelihood function to model the non-negative integer number of pumps. The model has conditions as fixed effects to estimate the differences between the two conditions. It models participants as random intercepts, and has trials as both fixed effects and random slope to account for learning effects. We provide code and model files in supplementary materials.

Not recommended: difficult to meaningfully report details and priors, reducing transparency

Transparency

Step 1 : Model specification in paper

Option 2. Mathematical equations

$$
\begin{array}{ll}
y_i & \sim \text{Neg-Binomial}(\mu, \Phi) \\
log(\mu) = \beta_{0,k} + \beta_{1,\, i} \,.\, x_i + \beta_{2,\, i} \,.\, x_j & \\
\beta_0 & \sim \text{Normal}(\alpha, \sigma) \\
\alpha & \sim \text{Normal}(4.1, 0.5) \\
\sigma & \sim \text{Exponential}(1) \\
\beta_{1,i} & \sim \text{Normal}(0, 0.5) \\
\Phi & \sim \text{Gamma}(5, 1) \\
i & \in \{\text{expansive, constrictive}\} \\
\text{j} & \in \{0, 1, 2, \ldots, n\}
\end{array}
$$

We model each response as a Negative Binomial distribution. The mean of this distribution varies with different conditions (β_1), trial number (β_2), and random intercepts for each participant ($\beta_{0,k}$).

Step 1 : Model specification in paper

Option 2. Mathematical equations

$y_i \quad \sim \text{Neg-Binomial}(\mu, \Phi)$

$log(\mu) = \beta_{0,k} + \beta_{1,\,i} \,.\, x_i + \beta_{2,\,i} \,.\, x_j$

$\beta_0 \quad \sim \text{Normal}(\alpha, \sigma)$

$\alpha \quad \sim \text{Normal}(4.1, 0.5)$

$\sigma \quad \sim \text{Exponential}(1)$

$\beta_{1,i} \quad \sim \text{Normal}(0, 0.5)$

$\Phi \quad \sim \text{Gamma}(5, 1)$

$i \quad \in \{\text{expansive, constrictive}\}$

$\text{j} \quad \in \{0, 1, 2, \ldots, n\}$

We model each response as a Negative Binomial distribution. The mean of this distribution varies with different conditions (β_1), trial number (β_2), and random intercepts for each participant ($\beta_{0,k}$).

Recommended if you are able to write out your model like this, and if you have space in your paper!

Transparency

Step 1 : Model specification in paper

Option 3. Use Wilkinson-Rogers-Pinheiro-Bates's notation

> We use a Negative Binomial likelihood function to model the non-negative integer number of pumps. In Wilkinson-Rogers-Pinheiro-Bates's notation, the model formula is
>
> $$pumps \sim condition + trial + (1 \mid participant)$$
>
> where condition is the two experiment conditions, trial is trial index to account for learning effects, and participants are modeled as random intercepts to account for correlation in the responses from the same participant.

Recommended if your model can be expressed in a syntax similar to Wilkinson's notation.

Transparency

Step 1 : Model specification in paper

Do not forget to report your priors!

You can either report them in the paper itself or in the supplement

If reporting priors in the supplement, link to them

Transparency

Step 2 :Model diagnostics in paper

```
summary(m)
```

```
Group-Level Effects:
~participant (Number of levels: 80)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.31      0.03     0.27     0.37 1.00     1407     2506

Population-Level Effects:
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept              3.52      0.05     3.42     3.63 1.00      724     1372
conditionexpansive    -0.02      0.07    -0.16     0.12 1.00      761     1902
trial                  0.01      0.00     0.00     0.01 1.00    12186     7888

Family Specific Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
shape     7.42      0.26     6.93     7.92 1.00    13088     9007
```

Step 2 :Model diagnostics in paper

> We check the minimum bulk (724) and tail (1,372) effective sample sizes to ensure reliable estimates, and check the maximum R-hat values (1.00) across all parameters to ensure model convergence.

Step 2 :Model diagnostics in paper

> We check the minimum bulk (724) and tail (1,372) effective sample sizes to ensure reliable estimates, and check the maximum R-hat values (1.00) across all parameters to ensure model convergence.

> We ran our model with 4 chains, each with 20,000 iterations. We discarded 10,000 warmup iterations per chain and thinned the final sample by 5, yielding 8,000 total post-warmup draws.

Step 2 :Model diagnostics in paper

> We check the minimum bulk (724) and tail (1,372) effective sample sizes to ensure reliable estimates, and check the maximum R-hat values (1.00) across all parameters to ensure model convergence.

> We ran our model with 4 chains, each with 20,000 iterations. We discarded 10,000 warmup iterations per chain and thinned the final sample by 5, yielding 8,000 total post-warmup draws.

Personally, we feel you don't need to report this in the paper

Step 3 : Reporting posteriors

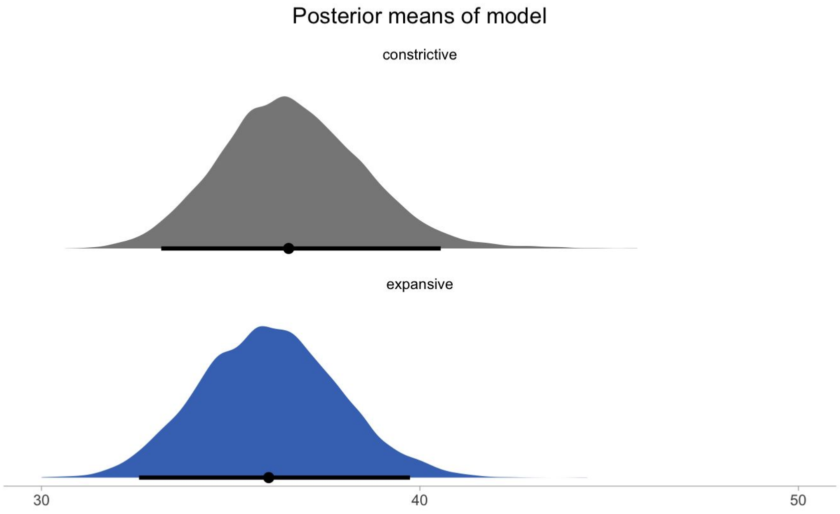
Guideline
Report those that answer your research questions. Inform readers on how to interpret the results.

> We use the average of all 30 trials and an average participant (random intercept = 0).

> We report medians and 95% Credible Intervals (CIs; Bayesian analog to confidence intervals) of posterior means. We also report the difference between the two conditions in the same format.

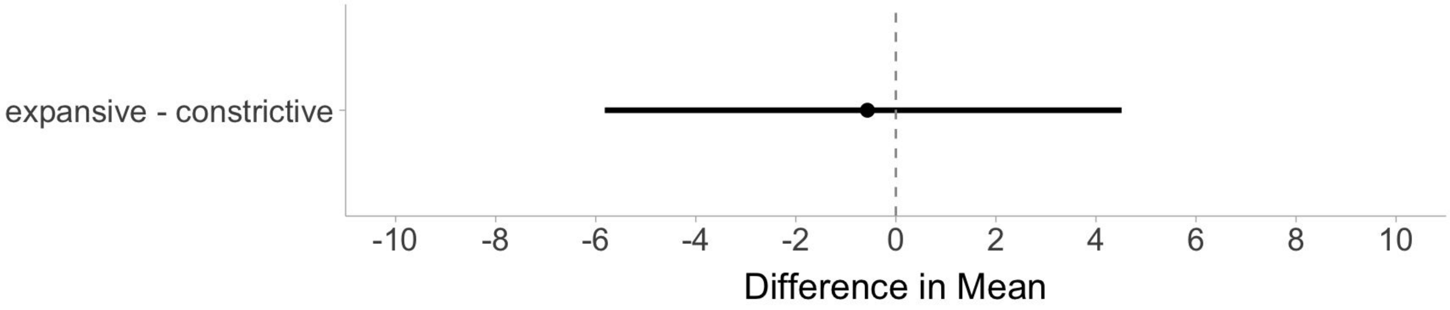
Step 3 : Reporting posteriors

For the average participant, the estimate of mean is 36.84 [33.33, 40.68] for the constrictive condition and 35.44 [32.08, 39.13] for the expansive condition.



Step 3 : Reporting posteriors

For the average participant, the difference of the mean between the two conditions is −0.56 [−5.82, 4.51], indicating no substantial difference.



Other things to report in the paper

Some reviewers may ask you to report the standard deviation of the random intercept. You can find them in the page printed out by `summary(...)`

```
## sd(Intercept)     0.31       0.03      0.27      0.37 1.00      1407      2506
```

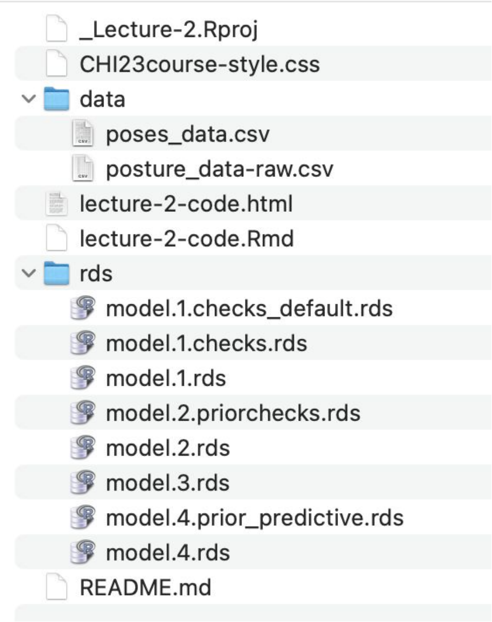
> The estimate of standard deviation for the random intercepts is 0.31 [0.27, 0.37].

Perhaps citing the libraries you use, **`brms`**, **`rstan`**, **`tidybayes`**, etc. in the paper.

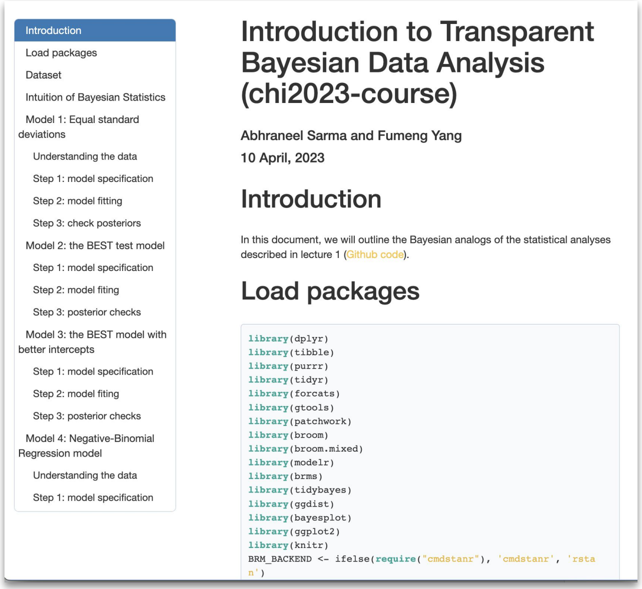
Perhaps version numbers, e.g., **`brms 2.15.0`**

Supplementary materials

Our course materials are an example.



The corresponding html



Introduction to Transparent Bayesian Data Analysis (chi2023-course)

Abhraneel Sarma and Fumeng Yang

10 April, 2023

Introduction

In this document, we will outline the Bayesian analogs of the statistical analyses described in lecture 1 (Github code).

Load packages

```
library(dplyr)
library(tibble)
library(purrr)
library(tidyr)
library(forcats)
library(gtools)
library(patchwork)
library(broom)
library(broom.mixed)
library(modelr)
library(brms)
library(tidybayes)
library(ggdist)
library(bayesplot)
library(ggplot2)
library(knitr)
BRM_BACKEND <- ifelse(require("cmdstanr"), 'cmdstanr', 'rsta
n')
```

Transparency

Supplementary materials

The `.Rmd` and its corresponding `.html`

- Likelihood, formula, priors, prior checks, posterior checks, etc.
- Other parameters: the number of chains, the number of iterations, etc.
- Perhaps the other models you tried
- Provide `sessionInfo()`

The model files `*.rds` (it may contain data)

Datasets (for ethical concerns, posteriors can be used as synthetic data)

A ReadMe to briefly explain your data and how to run the code

Anonymous review

Check the path that may contain your name(s)

Use a private OSF repository and provide a view-only link with anonymous authors

Limitations of Bayesian data analysis

Limitations of Bayesian data analysis

Computationally intensive

Priors require careful thought

Posterior-hacking

Unfamiliar reviewers might not like it

Computationally intensive

Complex Bayesian models can take hours or days. Here are some tips:

Get rid of columns you don't use to save memory
Try `CmdStanR` instead of `RStan` as backend
Remember to force a CPU core for each chain

Reparameterize (e.g., [0,1000]->[0, 1]) to make it easier for the sampler
Carefully think about the model (how complicated does it need to be)
Choose appropriate priors to help model converge

Faster CPUs (researchers are working on GPU implementation)

Plan early and take a break while the model is running

Posterior-hacking

P-hacking: trying different analysis methods (e.g., dropping data, different tests) to get a significant p value

Posterior-hacking: trying different priors and likelihoods to get desired posterior distributions

http://datacolada.org/13

Posterior-hacking

P-hacking: trying different analysis methods (e.g., dropping data, different tests) to get a significant p value

Posterior-hacking[1]: trying different priors and likelihoods to get desired posterior distributions

Pre-register details regarding your model including your priors
see: https://aspredicted.org/iv7jb.pdf

1. http://datacolada.org/13
2. Fernandes et al., "Uncertainty Displays Using Quantile Dotplots or CDFs Improve Transit Decision-Making" *Proceedings of the 2018 CHI conference on human factors in computing systems*. 2018.

Priors require careful thought

Think about

What parameter values are possible and how much probability mass the prior distribution assigns to them.

How the choice of priors impact the **prior predictive distribution**.

What the choice of priors imply about your results.

For practical issues in setting priors, see: Abhraneel Sarma and Matthew Kay. "Prior setting in practice: Strategies and rationales used in choosing prior distributions for Bayesian analysis." *Proceedings of the 2020 CHI conference on human factors in computing systems*. 2020.

Unfamiliar reviewers might not like it

Because statistical inference cannot be dichotomised to a *significant* or *insignificant* result, reviewers have complained that “results did not have strong statistical backing”.

Be careful to explain model construction and parameterization clearly, provide thoughtful models, supplementary materials, and detailed justifications.

Hope this course and your practice could help make a difference.

Other resources

What if I want to learn more?

Multilevel modeling

Model comparison and selection

Sensitivity and robustness analysis

Most can be found in the three books on the next slide.

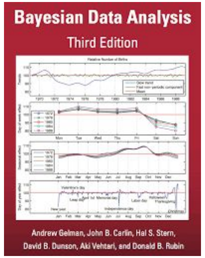
Preregistration—Lecture 1 by Chat Wacharamanotham

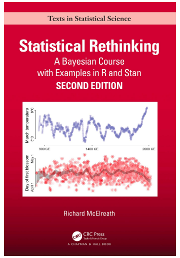
Visualizing the results—Lecture 3 by Xiaoying Pu on Monday

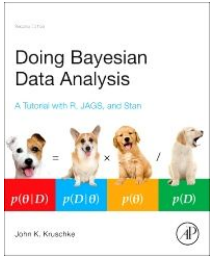
Resources

Books

Bayesian Data Analysis - Andrew Gelman et al. (2014)
Statistical Rethinking - Richard McElreath (2019, 2020)
Doing Bayesian Data Analysis - John K. Kruschke (2011)







Miscs

R library documents
Other people's code, Github
The Stan forum
Stan user manual

Online materials

https://bookdown.org/ajkurz/Statistical_Rethinking_recoded/
https://github.com/avehtari/BDA_course_Aalto
https://www.barelysignificant.com/slides/RGUG2019
https://www.sumsar.net/files/academia/user_2015_tutorial_bayesian_data_analysis_short_version.pdf
https://betanalpha.github.io/assets/case_studies/principled_bayesian_workflow.html
https://rss.onlinelibrary.wiley.com/doi/pdfdirect/10.1111/rssa.12378

Questions?

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Please answer a one-question survey about this session:
http://tiny.cc/t-chi23-feedback-2

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